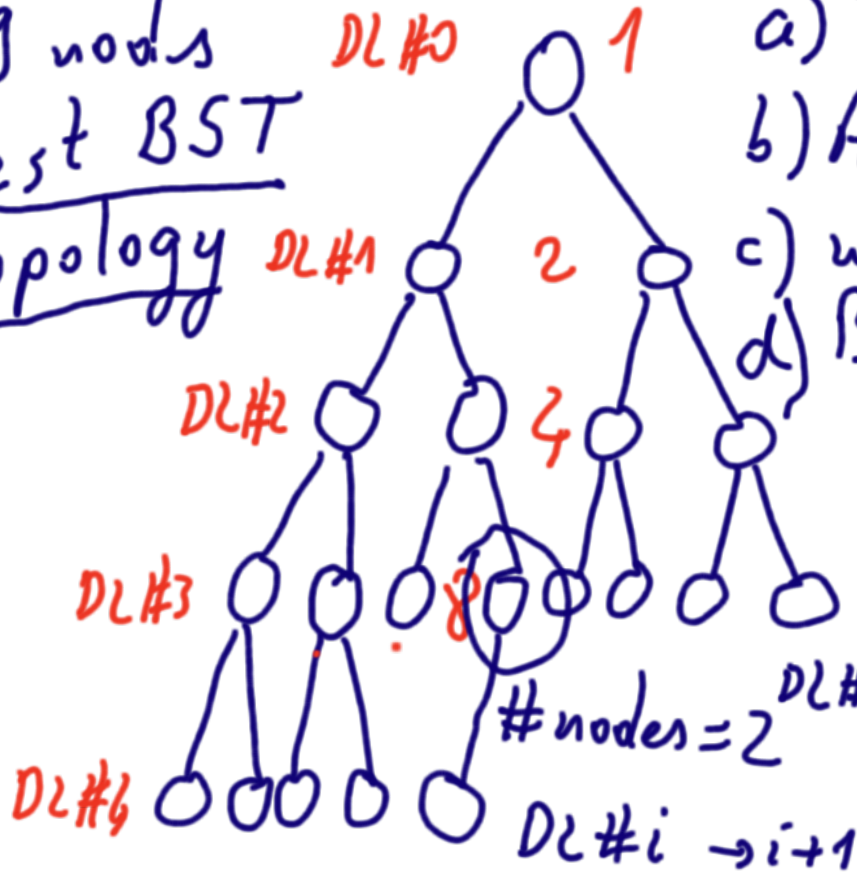
19 nodes

Best BST topology



a) Total access time
b) Average " "
c) Worst " "
d) Best " "

a) 69 "
b) 69/19
c) 5
d) 1

Q) n nodes in a BST with a best distribution of nodes
Provide the distribution of nodes at depth levels?

DL#	0	1	2	3	4	...	i	...	k	x
#of nodes	1	2	4	8	16	...	2^i	...	2^k	?

x: # nodes at the last DL

$x = n - (2^{k+1} - 1)$; $k = ?\, f(n)$ in terms of n only

e.g., for $\forall n \in \{16, \dots, 31\}$ $2^{k+1} - 1 = 15$

$\Rightarrow 2^{\lfloor \log_2 n \rfloor} - 1 \Rightarrow k+1 = \lfloor \log_2 n \rfloor$

$\Rightarrow x = n - [2^{\lfloor \log_2 n \rfloor} - 1]$